Linear Regression (Standard $\mathcal{D}$ = Batch)

Hyperparameters:

1) η (Learning Rate): For the following classical assumptions in Linear Regression, performed to find the best fit line;
$\rightarrow \mathcal{D} = \{(x^{(i)}, y^{(i)})\}_{i=1}^{m}$, m-samples and n-features per sample.
$\rightarrow$ We assume:

$$y^{(i)} = \Theta^T x^{(i)} + \epsilon^{(i)}, \text{ where } \epsilon^{(i)} \sim \mathcal{N}(0, \sigma^2)$$

$$\text{and, } \epsilon^{(i)} \sim \mathcal{N}(0, \sigma^2) \text{ is I.I.D. } \forall i \in [m].$$

$\rightarrow$ From this, we perform MLE for the $\Theta \in \mathbb{R}^n$ parameter, and find that;
"If Θ^* is the perfect/best-fit set of parameters, then we want-"

$$\Theta^* = \arg\min_{\Theta} \frac{1}{2m} \sum_{i=1}^{m} \left(\Theta^T x^{(i)} - y^{(i)}\right)^2 = \arg\min_{\Theta} \frac{1}{2m} (X\Theta - Y)^T (X\Theta - Y)$$

where $x^{(i)} \in \mathbb{R}^n$, $y^{(i)} \in \mathbb{R}$, $X \in \mathbb{R}^{m \times n}$ (sample features), $Y \in \mathbb{R}^{m \times 1}$ (sample targets).
Here, $\frac{1}{2m}(X\Theta - Y)^T(X\Theta - Y) = J(\Theta)$, also called the cost function.
$\rightarrow$ From this we get the update equation for Θ, as well as the analytical solution-

- $\Theta^*_{\text{analytical}} = \arg\min_{\Theta} \frac{1}{2m}(X\Theta - Y)^T(X\Theta - Y)$

$$\nabla_{\Theta} J(\Theta^*_{\text{analytical}}) = \frac{1}{m} X^T (X\Theta^*_{\text{analytical}} - Y) = 0$$

- $\Theta^{(t+1)} = \Theta^{(t)} - \eta \nabla_{\Theta} J(\Theta^{(t)})$
  $\Rightarrow$ this simply becomes;

$$\Theta^{(t+1)} = \Theta^{(t)} - \eta \left(\frac{1}{m}\right) X^T (X\Theta^{(t)} - Y)$$

$\Rightarrow$ Now, since we analyze this from an optimization-cum-learning framework, we answer the following questions:-

Q1. What is η? What value does the update recurrence rule converge to? Does it converge at all?

A1. $\eta \rightarrow$ learning rate: Simply determines how "fast" the recurrence relation changes the value of $\Theta^{(t)}$.
Our main goal, as we expect, is:

$$\lim_{t \to \infty} \left(\Theta^{(t+1)} = \Theta^{(t)} - \eta \left(\frac{1}{m}\right) X^T (X\Theta^{(t)} - Y) \right) \Rightarrow \Theta^*_{\text{analytical}}$$

Solving this, we have:

$$\Theta^{\infty} = \Theta^{\infty} - \eta \left(\frac{1}{m}\right) X^T(X\Theta^{\infty} - Y)$$

$$\Rightarrow \eta \left(\frac{1}{m}\right) X^T(X\Theta^{\infty} - Y) = 0 \quad \text{or} \quad \nabla_{\Theta} J(\Theta^{\infty}) = 0.$$

But since, $\nabla_{\Theta} J(\Theta^*_{\text{analytical}}) = 0$, hence $(\Theta^{\infty} = \Theta^*_{\text{analytical}})$.

- However, this is only possible if we assume Θ^{∞} is a finite value. How do we guarantee that?
- Let's say, $\Theta^{(t)} - \Theta^* = e^{(t)}$ $(\Theta^* = \Theta^*_{\text{analytical}})$. Be the error/distance, of how far $\Theta^{(t)}$ is currently, from Θ^*.

$$\Rightarrow (\Theta^{(t+1)} - \Theta^*) = (\Theta^{(t)} - \Theta^*) - \eta \left(\frac{1}{m}\right) X^T[X(\Theta^{(t)} - \Theta^*) + X\Theta^* - Y]$$

  This simplifies to:

$$e^{(t+1)} = e^{(t)} - \eta \left(\frac{1}{m}\right) X^T X \cdot e^{(t)}$$

$$\Rightarrow e^{(t+1)} = \left(I - \eta \left(\frac{1}{m}\right) X^T X\right) e^{(t)}$$

Now, our main goal is $(\|e^{(\infty)}\|_2^2 \to 0)$, i.e. sequence of errors converges to origin. For that, an important condition in terms of these scalar errors is -

$$\frac{\|e^{(t+1)}\|_2^2}{\|e^{(t)}\|_2^2} < 1$$

Using this condition, we further get-

$$\frac{\left\|\left(I - \left(\frac{\eta}{m}\right) X^T X\right) e^{(t)}\right\|_2^2}{\|e^{(t)}\|_2^2} < 1$$

$\rightarrow$ Let's call the LHS value as B.

We would want, that no matter what, $B < 1$. Thus, a guarantee we take is- $B_{\max} < 1$, since $(B \le B_{\max} \le 1)$.

Now, a famous inequality in terms of spectral norms;

$$\frac{\left\|\left(I - \eta \left(\frac{1}{m}\right) X^T X\right) e^{(t)}\right\|_2^2}{\|e^{(t)}\|_2^2} \le \left|\lambda_{\max}\left(I - \eta \left(\frac{1}{m}\right) X^T X\right)\right|^2 < 1$$

This would simply imply: if $H = \frac{1}{m} X^T X$,

$$-1 < \lambda_i(I - \eta H) < 1$$

Now, for our first bound:
$\rightarrow$ we want $B_{\max} < 1$:

$$\Rightarrow \lambda_{\max}(I - \eta H) = 1 - \eta\lambda_{\min}(H) < 1$$
$$\Rightarrow \eta\lambda_{\min}(H) > 0 \Rightarrow \eta > 0$$

Now, for second bound:
$\rightarrow$ we want $B_{\min} > -1$:

$$\Rightarrow \lambda_{\min}(I - \eta H) = 1 - \eta\lambda_{\max}(H) > -1$$
$$\Rightarrow 2 > \eta\lambda_{\max}(H)$$
$$\Rightarrow \frac{2}{\lambda_{\max}(H)} > \eta$$

Now, combining the 2 bounds, we get;

$$0 < \eta < \frac{2}{\lambda_{\max}\left(\frac{1}{m}X^T X\right)}$$

For this particular range of values, the recurrence relation converges to a finite value. For values outside of this range, we get -

$$\frac{\|e^{(t+1)}\|_2^2}{\|e^{(t)}\|_2^2} \geq 1$$

i.e. error either remains the same ($= 1$) or diverges (> 1).

Q2. If η is a hyperparameter, then what does it mean if we say an optimal η exists for regression best-fit problems? What is that value?

A2. In other words, this is simply;
$\eta_{\text{opt}} \rightarrow$ The value of η for which the sequence converges with least no. of iterations. This is different from *fastest*, since;

$$\eta_{\text{fast}} = \left(\frac{2}{\lambda_{\max}(H)}\right)^-$$

To find this value, we essentially want to do the following:

$$\arg\min_{\eta} N(\eta)$$

where $N(\eta) \rightarrow$ the no. of iterations it takes to converge to Θ^*.

Heuristically speaking, convergence $\neq$ equivalence. So, in essence, N always tends to ∞. So, what to do? We reframe the problem-
"We want to find $\arg\min_\eta N(\eta)$ where N is the no. of steps it takes for $\Theta^{(t)}$ to enter a ballpark of ε around Θ^*, where: $\varepsilon > 0$".

Thus, ($\lim_{\varepsilon\to 0} \Rightarrow \lim_{N\to\infty}$). For a finite but small ε, we will similarly have a finite but large N. Accordingly, Now, from our error update equation, we have -

$$e^{(N)} = \left(I - \eta\left(\frac{1}{m}X^T X\right)\right) e^{(N-1)}$$

If we set at $t = 0$, $e^{(0)}$ to be the error, then we get;

$$e^{(N)} = \left(I - \eta\left(\frac{1}{m}X^T X\right)\right)^N \cdot e^{(0)}$$

as a result of the recurrence.

Now, our goal is: $\|e^{(N)}\|_2 < \varepsilon$. OR,

$$\Rightarrow \frac{\|e^{(N)}\|_2}{\|e^{(0)}\|_2} < \left(\frac{\varepsilon}{\|e^{(0)}\|_2}\right)$$

We expect $\|e^{(0)}\|_2$ to be large, since upon initialization, we don't expect $\Theta^{(0)} = \Theta^*_{\text{analytical}}$.

Hence, we get:-

$$\frac{\left\|\left(I - \eta\left(\frac{1}{m}X^T X\right)\right)^N e^{(0)}\right\|_2}{\|e^{(0)}\|_2} < \frac{\varepsilon}{\|e^{(0)}\|_2}$$

This simply transforms into:

$$\left|\lambda_i\left(I - \eta\left(\frac{1}{m}\right)X^T X\right)\right|^N < \frac{\varepsilon}{\|e^{(0)}\|_2}$$

Now, since $\lambda_i(\eta)$ as a function of η, we do-

$$N < \frac{(\log(\varepsilon) - \log(\|e^{(0)}\|_2))}{\log|1 - \eta\lambda_i(H)|}$$

$$\Rightarrow N < f(\eta, \lambda_i)$$

Now, since $\log|1 - \eta\lambda_i|$ is convex in λ_i $\forall i \in [n]$, we can have 2 possible upper bounds for N;

1. $N < \frac{(\log(\varepsilon) - \log(\|e^{(0)}\|_2))}{\log|1 - \eta\lambda_{\max}(H)|}$ and
2. $N < \frac{(\log(\varepsilon) - \log(\|e^{(0)}\|_2))}{\log|1 - \eta\lambda_{\min}(H)|}$.

If we guarantee that N is smaller for *both* these bounds, and such that N itself is minimum, we will be able to get η_{opt}.

If we reframe this optimization problem:

$$\arg\min_\eta \left(\min_{\lambda_i \in \{\lambda_1 \ldots \lambda_n\}} \max f(\eta, \lambda_i)\right) = \eta_{\text{opt}}$$

Then, we essentially want -

$\arg\min_\eta f(\eta, \lambda_{\min})$ OR $\arg\min_\eta f(\eta, \lambda_{\max})$ to BOTH be η_{opt}. Thus,

$$f(\eta_{\text{opt}}, \lambda_{\min}) = f(\eta_{\text{opt}}, \lambda_{\max})$$

$$\Rightarrow \log|1-\eta\lambda_{\min}| = \log|1-\eta\lambda_{\max}|$$

Hence;
$\rightarrow 1-\eta\lambda_{\min} = -(1-\eta\lambda_{\max}) \Rightarrow \eta = 0$. [This violates convergence]
$\rightarrow -(1-\eta\lambda_{\min}) = 1-\eta\lambda_{\max}$

$$\Rightarrow \eta(\lambda_{\min}+\lambda_{\max}) = 2$$

$$\therefore \eta_{\text{opt}} = \frac{2}{\lambda_{\min}+\lambda_{\max}} \quad \forall(\varepsilon > 0)$$

Since this quantity is independent of ε, we can simply say:

$$\eta_{\text{opt}} = \frac{2}{\lambda_{\max}+\lambda_{\min}} \quad \forall\varepsilon \in \mathbb{R}^+$$

It is from this analysis, that we produce 3 learning rates, and test our linear regression model against them:-

- $\eta_1 \sim U\left(0, \frac{2}{\lambda_{\max}}\right)$. Guaranteed convergence. May not be fastest.
- $\eta_2 = \eta_{\text{opt}} = \left(\frac{2}{\lambda_{\max}+\lambda_{\min}}\right)$. Fastest convergence.
- $\eta_3 > \left(\frac{2}{\lambda_{\max}}\right)$. Guaranteed divergence.

* Analyze the loss-curves, and see expected results.

$\Rightarrow$ Now, for locally-weighted regression (LWR):-
<u>Assumptions</u>:-

$\rightarrow$ $\mathcal{D} = B$ (batch size per iteration).

$\rightarrow$ $\omega^{(i)} = \exp\left[-\dfrac{\|x^{(i)}-x_q\|_2^2}{2\tau^2}\right]$, where x_q is query, τ is hyperparameter.

$\rightarrow$ $J(\theta) = \left(\dfrac{1}{2m}\right)\sum_{i=1}^{m}\omega^{(i)}\left(\theta^T x^{(i)} - y^{(i)}\right)^2 = \dfrac{1}{2m}(X\theta-Y)^T W(X\theta-Y)$.

Here, $X \in \mathbb{R}^{m\times n}$, $Y \in \mathbb{R}^{m\times 1}$, $W \in \mathbb{R}^{m\times m}$ s.t. $W = \text{diag}(\omega^{(1)}, \omega^{(2)}, \ldots, \omega^{(m)})$,
$m =$ no. of samples, $n =$ feature per sample, $x^{(i)} \in \mathbb{R}^{n\times 1}$, $y^{(i)} \in \mathbb{R}$,
$\theta \in \mathbb{R}^{n\times 1}$, $\omega^{(i)} \in \mathbb{R}\ \ \forall i \in [m]$, $x_q \in \mathbb{R}^{n\times 1}$, $\tau \in \mathbb{R}^+$

$$\Rightarrow \nabla_\theta J(\theta) = \frac{1}{m}X^T W(X\theta - Y)$$

$$\Rightarrow \nabla_\theta^2 J(\theta) = \frac{1}{m}X^T W X = H(\theta) \quad \text{[Hessian for loss landscape].}$$

* Hyperparameters :-

1) η (learning rate): From results of linear regression, we see:

- $\cdot$ η has same bounds : $\boxed{0 < \eta < \dfrac{2}{\lambda_{\max}\left[\left(\frac{1}{m}\right) X^T W X\right]}}$
- $\cdot$ $\boxed{\eta_{\text{opt}} = \dfrac{2}{\lambda_{\max}(H) + \lambda_{\min}(H)}}$, considering the update equation to be the same;

$$\boxed{\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_\theta J(\theta^{(t)})}$$

However, some questions arise-

Q1. Is it possible to get an η that guarantees convergence, without calculating $H(\theta)$ at all?

A1. Absolutely! We demonstrate that here-

Let's consider the hessian $H(\theta)$. Upon looking at each individual term, we see that

$$[H_{ij}] = \left(\frac{1}{m}\sum_{p=1}^{m} \omega^{(p)}(x_i^{(p)})(x_j^{(p)})\right), \text{ where } H \in \mathbb{R}^{n\times n}$$

Then, let's define:

$$(X^T X)^{(p)} \Rightarrow \text{every element } [(X^T X)^{(p)}]_{ij} = (x_i^{(p)} \cdot x_j^{(p)}), \text{ where } (X^T X)^{(p)} \in \mathbb{R}^{n\times n}$$

* Then, we can define H through a summation (weighted) of these very matrices; (with an additional factor of $\frac{1}{m}$).

$$\Rightarrow H(\theta) = \frac{\omega^{(1)}(X^T X)^{(1)}}{m} + \frac{\omega^{(2)}(X^T X)^{(2)}}{m} + \cdots = \sum_{p=1}^{m} \frac{(\omega^{(p)}(X^T X)^{(p)})}{m}$$

Further observing, $(X^T X)^{(p)} = x^{(p)}(x^{(p)})^T \Rightarrow$ these are just outer products of each sample.

* Outer products have a special property:

$\boxed{\lambda_{\max}\left(x^{(k)}(x^{(k)})^T\right) = \|x^{(k)}\|_2^2}$. We will be using this prop. after this step!

$$H = \sum_{p=1}^{m} \frac{\left(\omega^{(p)} \cdot x^{(p)} \cdot (x^{(p)})^T\right)}{m} \cdot$$ Now, via the property of eigenvalue summation, we have;

$$\Rightarrow \lambda_{\max}(H) \leq \frac{1}{m}\sum_{p=1}^{m} \omega^{(p)} \cdot \lambda_{\max}\left[x^{(p)}(x^{(p)})^T\right]$$

$$\Rightarrow \left(\frac{2}{\lambda_{\max}(H)}\right) \geq \frac{m}{\left(\sum_{p=1}^{m} \omega^{(p)} \|x^{(p)}\|_2^2\right)} = \eta_{\text{guarantee}}$$

Since $\eta_{\text{guarantee}} > 0$, it will always lie inside our convergence bounds.

But, how does this benefit us?

Q2. Why is $\eta_{\text{guarantee}}$ faster to calculate then η_{opt} in some cases?

A2.

1. $\eta_{\text{guarantee}}$ takes:
   - $\rightarrow$ $x^{(i)}$, squares each element & adds those elements to get $\|x^{(i)}\|_2^2$: $\Theta(n)$.
   - $\rightarrow$ Multiply each $x^{(i)}$ with $\omega^{(i)}$: $\Theta(1)$
   - $\rightarrow$ Do this $\forall i \in [m]$; $\underbrace{\Theta(n) + \cdots + \Theta(n)}_{m \text{ times}} = \Theta(mn)$.

2. η_{opt} takes:
   - $\rightarrow$ finding Hessian : $\Theta(mn^2)$
   - $\rightarrow$ Calculating its eigenvalues
     - Worst case; $\Theta(n^3)$
     - Best case; $\Theta(n^2)$
   - $\rightarrow$ Net operation $= \Theta(mn^2 \text{ or } n^3)$, atleast!

I. For cases where $m \gg n$, 2 is easier to calculate.

II. For cases where $n \gg m$, 1 is easier to calculate, although we sacrifice some amount of convergence speed.

In our loss curves, we see :

1) $\eta_{\text{opt}} \longrightarrow$ Converges

2) $\eta_{\text{guarantee}} \longrightarrow$ Converges

3) $\eta_{\text{bounded}} \sim U\left(0, \frac{2}{\lambda_{\max}}\right) \longrightarrow$ Converges

4) $\eta_{\text{diverge}} > \left(\frac{2}{\lambda_{\max}}\right) \longrightarrow$ Diverges

* Thus, our mathematical behaviour for $\eta_{\text{guarantee}}$ successfully is validated by its empirical behaviour over LWR.